Within the flat 6D Conformal Spacetime Algebra system ($Cl(4,1,1)$), the complementary junction field-effect transistors—the N-channel **2N5458** and the P-channel **2N5460**—are modeled as symmetric, charge-gated conduction channels.

Rather than relying on quantum carrier wave-functions, their transport kinetics, transconductance curves, and saturation limits are derived from the classical depletion of mobile soliton channels and the drift of positive vacancies within the 3D silicon lattice.

Section 21: Complementary Junction Field-Effect Transistors (2N5458 and 2N5460)

The 2N5458 and 2N5460 form a complementary JFET pair. The 2N5458 modulates a channel of mobile electron solitons [4.1] using positive vacancy gates, while the 2N5460 modulates a channel of positive core vacancies (holes [5.2]) using electron soliton gates.

```
      [ 2N5458 N-Channel JFET ]                    [ 2N5460 P-Channel JFET ]
  Gate: P-type (Positive Vacancies)            Gate: N-type (Electron Solitons)
  Channel: N-type (Electron Solitons)          Channel: P-type (Positive Vacancies)
  Bias: V_GS < 0, V_DS > 0                     Bias: V_GS > 0, V_DS < 0
  Conduction: Electron Soliton Drift           Conduction: Positive Vacancy Drift
```

21.1 2N5458 N-Channel JFET Soliton Dynamics The 2N5458 consists of a continuous conducting channel of width W_0 in the x-direction, length L in the y-direction, and depth Z in the z-direction, carrying a donor doping density N_d of mobile electron solitons.

21.1.1 Electrostatic Depletion Width Applying a negative Gate-to-Source potential ($V_{GS} < 0$) establishes a perpendicular electric field E_{gate} across the gate-channel junction. In the static limit, this field satisfies the inhomogeneous master field equation [2.4]:

$$\nabla \cdot E_{\text{gate}} = \frac{\rho_e}{\epsilon_{\text{si}}}$$

This field repels the mobile electron solitons away from the Gate regions, creating a depletion region of local width $W_d(y)$ on each side of the channel:

$$W_d(y) = \sqrt{\frac{2\epsilon_{\text{si}}}{eN_d}\left[\Phi_0 - (V_{GS} - V(y))\right]}$$

where: * Φ_0 is the built-in junction potential. * $V(y)$ is the local channel potential at position y along the channel length.

The remaining undepleted channel width is $W(y) = W_0 - 2W_d(y)$.

21.1.2 Convective Soliton Drift Integration The current I_D of electron solitons drifting under the lateral electric field $E_{\text{lateral}} = -\frac{dV}{dy}$ is:

$$I_D = eN_d W(y) Z u_{\text{drift},n} = eN_d Z \mu_n \left[W_0 - 2\sqrt{\frac{2\epsilon_{\text{si}}}{eN_d}(\Phi_0 - V_{GS} + V(y))} \right] \frac{dV}{dy}$$

Integrating this relation along the channel length from $y = 0$ (where $V(0) = 0$) to $y = L$ (where $V(L) = V_{DS}$):

$$\int_0^L I_D \, dy = eN_d Z \mu_n \int_0^{V_{DS}} \left[W_0 - 2\sqrt{\frac{2\epsilon_{\text{si}}}{eN_d}(\Phi_0 - V_{GS} + V)} \right] dV$$

$$I_D = G_{0,n} \left[V_{DS} - \frac{2}{3} \frac{(V_{DS} - V_{GS} + \Phi_0)^{3/2} - (-V_{GS} + \Phi_0)^{3/2}}{V_{P,n}^{1/2}} \right] \quad \text{(Triode Region)}$$

where: * $G_{0,n} = \frac{eN_d W_0 Z \mu_n}{L}$ is the baseline channel conductance. * $V_{P,n} = \frac{eN_d W_0^2}{8\epsilon_{\text{si}}}$ is the negative pinch-off voltage threshold.

When $V_{DS} \geq V_{GS} - V_{P,n}$, the depletion regions merge at the drain end, pinching off the channel. The current saturates to a value governed strictly by the gate potential:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{P,n}} \right)^2 \quad \text{(Saturation Region)}$$

For the **2N5458**, the typical parameters are: * **Saturated Drain Current (I_{DSS}):** 2.0 mA to 9.0 mA at $V_{DS} = 15$ V, $V_{GS} = 0$. * **Gate-Source Cutoff Voltage ($V_{P,n}$):** -1.0 V to -7.0 V.

21.2 2N5460 P-Channel JFET Vacancy Dynamics The 2N5460 consists of a continuous conducting channel of P-type silicon containing an acceptor doping density N_a of localized positive core vacancies.

21.2.1 Electrostatic Depletion Width Applying a positive Gate-to-Source potential ($V_{GS} > 0$) generates a perpendicular electric field E_{gate} pointing into the channel, repelling the positive core vacancies and creating a depletion region of local width $W_{d,p}(y)$:

$$W_{d,p}(y) = \sqrt{\frac{2\epsilon_{\text{si}}}{eN_a}\left[\Phi_0 + (V_{GS} - V(y))\right]}$$

Because the channel potential $V(y)$ is negative ($V(y) < 0$) for a P-channel device, the term $V_{GS} - V(y)$ increases toward the drain, narrowing the conducting vacancy path.

21.2.2 Convective Vacancy Drift Integration Under a negative drain-to-source potential $V_{DS} < 0$, the positive vacancies drift from Source to Drain with mobility μ_p. Integrating the vacancy current along the channel length:

$$I_D = -G_{0,p}\left[|V_{DS}| - \frac{2}{3}\frac{(|V_{DS}| + V_{GS} + \Phi_0)^{3/2} - (V_{GS} + \Phi_0)^{3/2}}{V_{P,p}^{1/2}}\right] \quad \text{(Triode Region)}$$

where: * $G_{0,p} = \frac{eN_aW_0Z\mu_p}{L}$ is the baseline vacancy conductance. * $V_{P,p} = \frac{eN_aW_0^2}{8\epsilon_{\text{si}}}$ is the positive pinch-off voltage threshold.

When $|V_{DS}| \geq V_{GS} + V_{P,p}$, the channel pinches off at the drain end, and the current saturates:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_{P,p}}\right)^2 \quad \text{(Saturation Region)}$$

For the **2N5460**, the typical parameters are: * **Saturated Drain Current (I_{DSS}):** -1.0 mA to -5.0 mA at $V_{DS} = -15$ V, $V_{GS} = 0$. * **Gate-Source Cutoff Voltage ($V_{P,p}$):** $+1.0$ V to $+6.0$ V.

21.3 Mobilities and Parameter Comparison Because positive core vacancies (holes) have lower mobility in the silicon lattice than mobile electron solitons ($\mu_p \approx 450\ \text{cm}^2/\text{V}\cdot\text{s}$ versus $\mu_n \approx 1400\ \text{cm}^2/\text{V}\cdot\text{s}$ [20.4]):

- **Conductance Ratio:** For identical channel geometries (W_0, L, Z) and doping densities ($N_d = N_a$), the baseline vacancy conductance of the 2N5460 is significantly lower than the baseline soliton conductance of the 2N5458:
$$\frac{G_{0,p}}{G_{0,n}} = \frac{\mu_p}{\mu_n} \approx 0.32$$

- **Transconductance Modulation:** The forward transconductance g_{fs} represents the sensitivity of the output current to the transverse gate electric field [21.4]:

$$g_{fs} = \frac{\partial I_D}{\partial V_{GS}} = \frac{2I_{DSS}}{|V_P|}\left(1 - \frac{V_{GS}}{V_P}\right)$$

  - For the N-channel **2N5458**, the maximum transconductance at $V_{GS} = 0$ is typically 4000 μS (4.0 mS).
  - For the P-channel **2N5460**, the maximum transconductance at $V_{GS} = 0$ is typically 2500 μS (2.5 mS).

This lower transconductance of the 2N5460 is a direct consequence of the lower drift velocity of the positive vacancies under the same lateral electric field gradient, confirming that the physical conduction properties of these complementary devices are governed by the different transport velocities of their charge-carrying solitons.